

A GUI for Magboltz

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WG4 — Modeling and Simulations

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Magboltz: Context & Workflow

Why it matters

- ▶ Monte Carlo electron transport simulation
- ▶ Fundamental tool for detector physics
- ▶ Used across MPGDs, TPCs, RPC studies

CLI friction

- ▶ Text-based input cards
- ▶ Manual output parsing
- ▶ Iterative tuning & plotting scripts
- ▶ Student onboarding difficulty

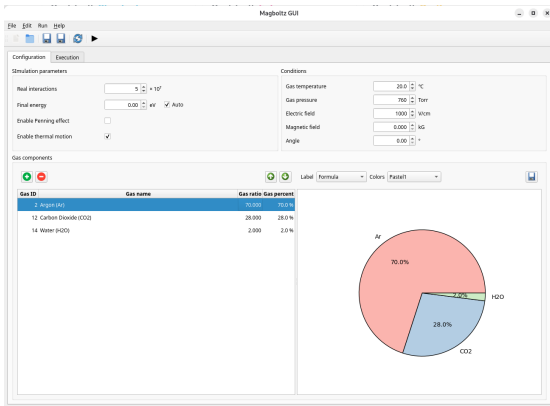
Cornerstone physics engine, workflow-heavy interface.

Input → Run → Parse → Export → Plot

We change the workflow, not the physics.

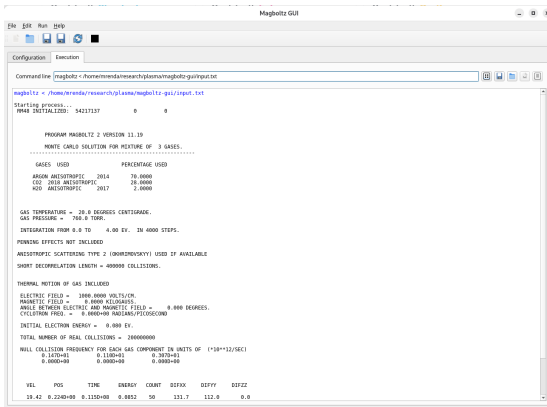
Magboltz GUI

- ▶ Python + PyQt6 application
- ▶ Cross-platform
- ▶ Designed as a workflow layer, not a replacement
- ▶ Core functions:
 - ▶ Input file creation & editing
 - ▶ Gas mixture editor & visual pie chart
 - ▶ Magboltz execution wrapper
 - ▶ Output parsing → structured model
 - ▶ Data export → CSV / JSON / XML
 - ▶ Built-in plotting tools



Execution Workflow

- ▶ Direct Magboltz invocation
- ▶ Command line transparency
- ▶ Live stdout/stderr streaming
- ▶ Run/Stop controls
- ▶ Raw output persistence
- ▶ Reopen previous runs



The screenshot shows the Magboltz GUI window. The 'Configuration' tab is active, displaying the command line: `magboltz < /home/mrenda/research/plasma/magboltz-gui/input.txt`. The 'Execution' tab is also visible, showing the output of the program. The output includes the version number (11.19), the Monte Carlo solution for a mixture of 3 gases, and a table of gases used. The table lists Argon Anisotropic (2014, 70.0000%), CO2 2018 Anisotropic (20.0000%), and R20 Anisotropic (2017, 2.0000%). The output also includes parameters such as gas temperature (20.0 degrees Centigrade), gas pressure (760.0 Torr), integration range (0.0 to 4.00 EV), and initial electron energy (0.000 EV).

```
Command line: magboltz < /home/mrenda/research/plasma/magboltz-gui/input.txt

magboltz < /home/mrenda/research/plasma/magboltz-gui/input.txt
Starting process...
PNAME INITIALID: 942117137 0 0

PROGRAM MAGBOLTZ 2 VERSION 11.19
MONTE CARLO SOLUTION FOR MIXTURE OF 3 GASES.
*****
GASES USED PERCENTAGE USED
ARGON ANISOTROPIC 2014 70.0000
CO2 2018 ANISOTROPIC 20.0000
R20 ANISOTROPIC 2017 2.0000

GAS TEMPERATURE = 20.0 DEGREES CENTIGRADE.
GAS PRESSURE = 760.0 TORR.
INTEGRATION FROM 0.0 TO 4.00 EV. IN 4000 STEPS.
PIONISING EFFECTS NOT INCLUDED
ANISOTROPIC SCATTERING TYPE 2 (BORSHCHINSKY) USED IF AVAILABLE
SHORT DECORRELATION LENGTH = 400000 COLLISIONS.

THERMAL MOTION OF GAS INCLUDED
ELECTRIC FIELD = 1800.0000 VOLTS/CM.
MAGNETIC FIELD = 0.0000 KILOGAUSS.
ANGLE BETWEEN ELECTRIC AND MAGNETIC FIELD = 0.000 DEGREES.
CYCLOTRON FREQ. = 0.000000 RAD/MS/PERIODSECOND
INITIAL ELECTRON ENERGY = 0.000 EV.
TOTAL NUMBER OF REAL COLLISIONS = 200000000
NULL COLLISION FREQUENCY FOR EACH GAS COMPONENT IN UNITS OF (*10**12/SEC)
0.1400+01 0.1100+01 0.3070+01
0.0000+00 0.0000+00 0.0000+00

VEL POS TIME ENERGY COUNT DIFX DIFY DIFZZ
18.42 0.2240+00 0.1150+00 0.8852 50 131.7 112.0 0.0
```

What This Does NOT Do

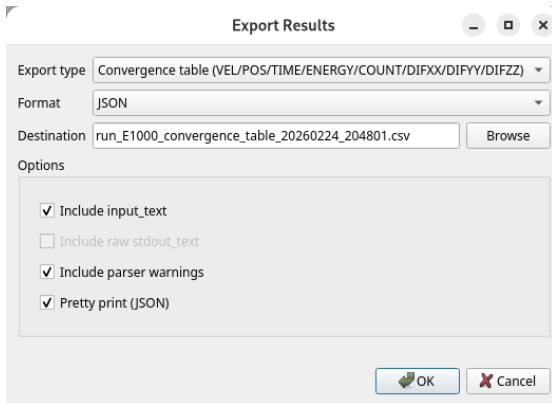
- ▶ Does **not** change Magboltz physics, models, or algorithms
- ▶ Does **not** alter input semantics or units
- ▶ Does **not** replace the CLI (it wraps it)
- ▶ Does **not** ship the Magboltz binary

Validation & Trust

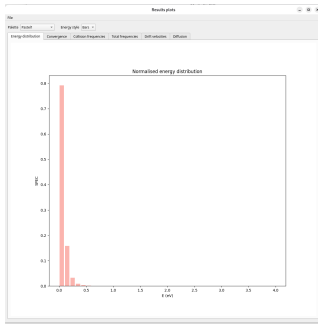
- ▶ Parsed outputs are **1:1** with Magboltz text output
- ▶ No unit conversions; values are preserved as printed
- ▶ Export schema is versioned (v1.0)
- ▶ Raw output & parser warnings are retained

From Text Output → Structured Data

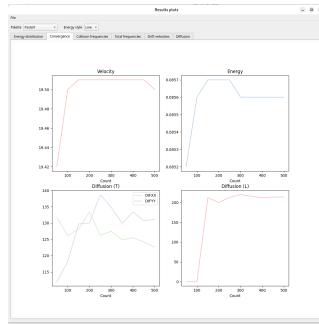
- ▶ Canonical RunResult model
- ▶ Extracted:
 - ▶ Transport coefficients
 - ▶ Collision frequencies
 - ▶ Convergence tables
 - ▶ Energy distributions
- ▶ Export formats:
 - ▶ CSV
 - ▶ JSON
 - ▶ XML
- ▶ Reproducible analysis pipelines



Built-in Plotting Tools



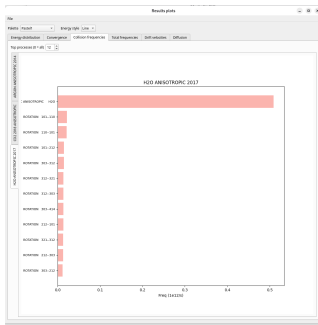
Energy distribution



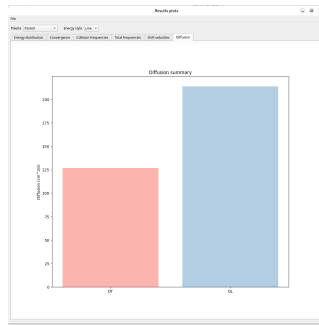
Convergence

Immediate visual feedback without external scripts.

Built-in Plotting Tools (cont.)



Collision frequencies



Diffusion summary

Immediate visual feedback without external scripts.

Project repository

- ▶ GitLab: <https://gitlab.com/micrenda/magboltz-gui/>
- ▶ License: MIT
- ▶ Documentation: README in the repository

Setup: pip (bundled Qt)

Recommended when you want Qt bundled with the app.

- ▶ Python environment
- ▶ Magboltz binary available in PATH

Commands:

```
pip install "magboltz-gui[qt]"  
magboltz-gui
```

Setup: pip (system Qt)

Recommended on Linux when using system PyQt6.

- ▶ Install system Qt bindings
- ▶ Magboltz binary available in PATH

Commands (Debian/Ubuntu):

```
sudo apt install python3-pyqt6  
pip install magboltz-gui  
magboltz-gui
```

Limitations & Next Steps

- ▶ No batch sweeps yet; single-run workflow
- ▶ Limited automation for parameter scans
- ▶ Next: parameter scan runner + input templates
- ▶ Next: headless export-only mode for pipelines

Stephen F. Biagi

(1949–2025)

We dedicate this work to his memory.

His pioneering contributions to modeling
electron transport in gases profoundly shaped
detector physics.

This interface is a modest tribute to his legacy.